

SESSION 6 — DIMENSIONAL MODEL DESIGN

Milestone B3 · Grain, facts and dimensions, schema on paper

BDAT04 Fundamentals of Data Warehousing · Saturday 26 September 2026 · Three hours

How this activity works

- Eight items, in three parts. Work them in order — each one uses what the one before it produced.
- Every item has the same two boxes. ANALYSIS is what you found or built. REFLECTION is what it means. Both are marked, and the reflection is worth more than you expect.
- You may use Power BI, your notes, your earlier queries and your feedback letters. Part C is done on paper.
- This activity IS milestone B3. There is no separate submission.

A reflection is not a summary.

Do not tell me what you did — the analysis box already says that. Tell me what it cost, what it would have broken, or what you now see that you did not see before. Two or three honest sentences beat a paragraph of restatement.

Dataset: Kape Tayo, as it stands in your own Power BI workspace. Nothing new is issued today.

PART A — GRAIN

Grain is the single sentence that says what one row of a fact table means. It is the first thing a designer declares and the most expensive thing to get wrong.

ITEM 1 • What is one row of tblSales?

about 20 minutes

What to do

1. Load tblSales. Count the total rows.
2. Count the DISTINCT values of SaleID.
3. Compare the two numbers. If they are equal, each SaleID appears once. If they differ, some appear more than once.
4. Look at three actual rows. What does each one describe — a whole sale? a single product within a sale? a day of sales?

ANALYSIS

State the grain in ONE sentence, beginning exactly: "One row of tblSales is ..." Then give the two counts you used.

One row of tblSales is a single item within a sale, recorded with its SaleID, Date, Branch, Product, etc.

Total rows: 10,589

Distinct values: 10,566

The two counts are different. This means some SaleID values appear more than once. After checking actual rows, the rows show information about a specific product or item sold, such as the product, quantity, price, and amount.

REFLECTION

If your grain sentence is wrong, every number built on this table is wrong and nothing warns you. Which of the two counts would have exposed the error — and what would you have concluded if you had only looked at the rows?

The distinct SaleID count would have shown the error. If I only looked at the rows, I might think that each row was one whole sale because every row has a SaleID and other sale information. But there are 10,589 rows and only 10,566 different SaleIDs, so some SaleIDs are repeated. This shows that one row should not automatically be treated as one whole sale.

ITEM 2 • What is one row of tblInventory?

about 20 minutes

What to do

5. Load tblInventory. Count the rows.
6. List the distinct values of MovementType and count the rows in each.
7. Decide what a single row records. Is it a stock level, or a stock movement? Those are not the same thing.
8. Check whether any column could serve as a unique identifier for the row.

ANALYSIS

State the grain in ONE sentence, beginning “One row of tblInventory is ...” List the MovementType values with their counts.

One row of tblInventory is one stock movement for a product at a branch on a specific date.

Total rows: 1,541

Delivery: 1,451

delivery: 25

Wastage: 23

Adjustment: 23

Transfer Out: 19

MovementID can be used as a unique identifier because there are 1,541 different MovementID values for 1,541 rows.

REFLECTION

tblSales and tblInventory are both fact tables, and their grains are different. Write one sentence explaining why you cannot put them in the same table, even though both are about products at branches over time.

tblSales and tblInventory are both fact tables, but their grains are different, so they cannot be placed in the same table because one row in Sales represents a product sold, while one row in Inventory represents a stock movement.

ITEM 3 • The grain test

about 25 minutes

What to do

9. Take this business question: “How much revenue did each branch earn in March 2026?” Using only your declared grain for tblSales, decide whether the table can answer it. Show the grouping you would use.
10. Now take this one: “What was the average number of items per customer visit?” Decide whether your declared grain can answer it.
11. Do not build the answer. Decide whether the grain permits it, and show why.

ANALYSIS

For each question: ANSWERABLE or NOT ANSWERABLE, and the one-line reason that follows from your grain statement.

How much revenue did each branch earn in March 2026?

ANSWERABLE

The table has the sale amount, branch, and date, so I can group Amount by BranchID for March 2026.

Grouping I would use: Grouping: BranchID + March 2026 → SUM(Amount)

What was the average number of items per customer visit?

NOT ANSWERABLE

The table has products and customers, but it does not have a column that clearly identifies a customer visit. The grain is one product or item in a sale, not one customer visit.

REFLECTION

One of these two questions is blocked by the grain, not by missing data. Explain the difference between “we do not have that data” and “the grain of this table cannot express that” — and say which one a business user would find harder to accept.

“We do not have that data” means the needed information is missing from the table. “The grain cannot express that” means the table has data, but the way each row is recorded does not represent the thing we are trying to measure. A business user would probably find the second one harder to accept because the data is there, but it is recorded at a different level.

PART B — FACTS AND DIMENSIONS

A fact table holds the things you measure. A dimension table holds the things you slice by. Most modelling errors are really classification errors made early and never revisited.

ITEM 4 • Classify every column in tblSales

about 25 minutes

What to do

12. List every column in tblSales.
13. Put each one into exactly one of three buckets: MEASURE (something you would sum or average), KEY (something that points at another table), DESCRIPTIVE ATTRIBUTE (something you would filter or group by, but never sum).
14. Two columns will be arguable. Decide anyway, and mark them with an asterisk.

ANALYSIS

A three-column table: Column | Bucket | Why. Every column in the table must appear exactly once.

Column	Bucket	Why
SaleID*	DESCRIPTIVE ATTRIBUTE	Identifies the sale but is not something to add or average.
Date	DESCRIPTIVE ATTRIBUTE	Used to group sales by date or time.
BranchID	KEY	Connects the sale to the Branches table.
ProductID	KEY	Connects the sale to the Products table.
CustomerID	KEY	Connects the sale to the Customers table.
Quantity	MEASURE	Shows how many items were sold and can be added.
UnitPrice*	MEASURE	Shows the price of one item and can be used to calculate averages or other price measures.
Amount	MEASURE	Shows the sale amount and can be added to get total revenue.
PaymentMethod	DESCRIPTIVE ATTRIBUTE	Used to group or filter sales by payment method.

REFLECTION

Find a column you could technically sum but never should. Name it and say what the sum would mean — and why producing that number would mislead a reader.

SaleID could technically be added, but it should never be summed. Adding SaleIDs would only produce a large number with no useful meaning, and it could make the reader think the number represents sales or revenue.

ITEM 5 · Classify the seven tables

about 25 minutes

What to do

- 15. List the seven Kape Tayo tables.
- 16. Classify each as FACT, DIMENSION, or NEITHER.
- 17. For each FACT, write its grain in one line (you already have two of them).
- 18. For each DIMENSION, name the column that joins it to a fact table.

ANALYSIS

A four-column table: Table | Fact / Dimension / Neither | Grain (facts only) | Join key (dimensions only).

Table	Fact / Dimension / Neither	Grain (facts only)	Join key (dimensions only)
tblSales	FACT	One row is one product or item in a sale.	
tblInventory	FACT	One row is one stock movement for a product at a branch on a date.	
tblCustomers	DIMENSION		CustomerID
tblBranches	DIMENSION		BranchID
tblCalendar	DIMENSION		Date
tblProducts	DIMENSION		ProductID
tblEmployees	NEITHER		

REFLECTION

Name the table you found hardest to classify and say what made it hard. A hard case is not a failure — it usually means the business itself has not decided something.

The hardest table to classify was tblEmployees because it has useful employee information, but there is no EmployeeID in tblSales or tblInventory to connect it to a fact table.

ITEM 6 • The hard case — tblEmployees

about 20 minutes

What to do

19. Open tblEmployees. Read its columns.
20. Ask: which fact table has a column that points at an employee? Check tblSales and tblInventory.
21. Decide: is tblEmployees a dimension in THIS model, or is it a table that belongs to a different model entirely?
22. If it is a dimension, name the fact table it would serve and the key it would use. If it is not, say what it would take to make it one.

ANALYSIS

Your decision, in writing, with the evidence you checked.

tblEmployees is not a dimension in this model. I checked tblSales and tblInventory, and neither table has an EmployeeID column. The employee table has employee information such as EmployeeID, EmployeeName, BranchID, Role, HireDate, DailyRate, and EmploymentStatus, but there is no employee key in the fact tables to connect it.

To make it a dimension, the fact table would need an EmployeeID that connects each sale or inventory movement to an employee.

REFLECTION

A dimension nobody joins to is not a dimension — it is a table sitting in your file. What would you say to a manager who insists it belongs in the model because “we have the data”?

I would tell the manager that having the data does not automatically mean it belongs in the model. The employee table needs a connection to a fact table. Without an EmployeeID in Sales or Inventory, the employee table cannot be used as a dimension in this model.

PART C — SCHEMA ON PAPER

You draw the schema before you build it, because paper forces you to decide what the software would quietly decide for you.

ITEM 7 • Draw the schema on paper

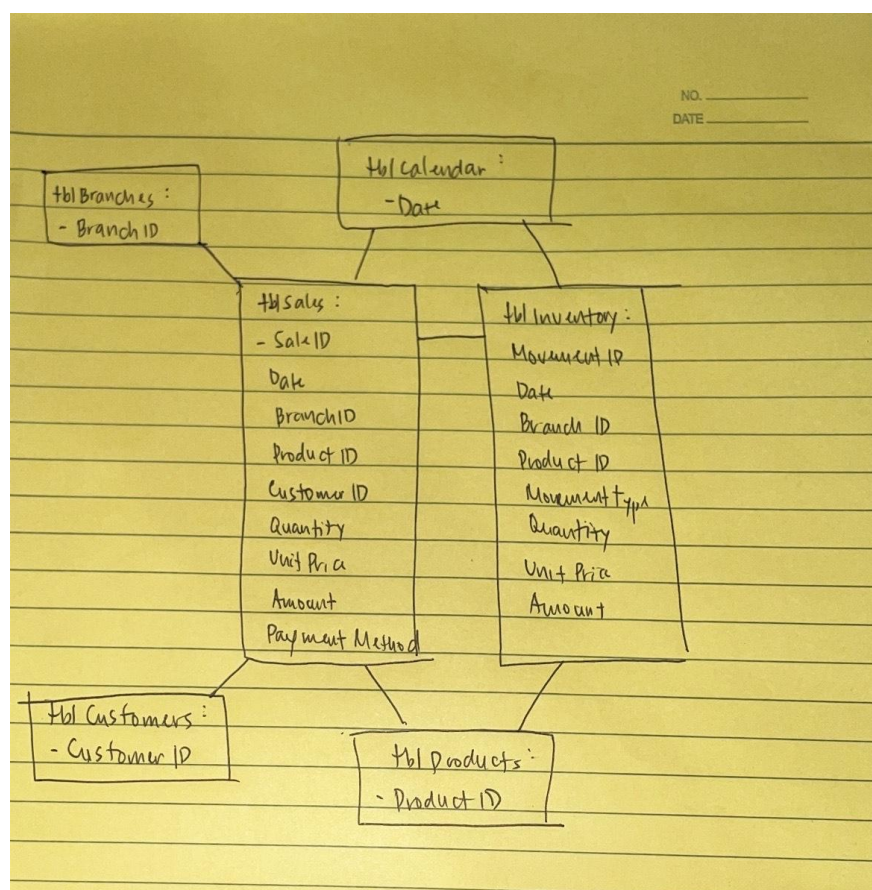
about 30 minutes

What to do

23. On PAPER — not in Power BI — draw your model.
24. Put your fact table or tables in the centre. Put the dimensions around them.
25. Draw a line for every relationship. Write the key name on each line.
26. Mark the SIDE of each relationship: which table is ONE, which is MANY. Use a 1 and a *.
27. Label your diagram with your name and photograph it.

ANALYSIS

A photograph of your drawing, inserted below or submitted alongside this sheet.



REFLECTION

You drew this before building it. Why does the syllabus ask for paper first? Answer from what happened while you were drawing — something you had to decide on paper that the software would have decided for you.

The paper drawing made me decide which tables should connect to each fact table and which side should be one or many. I then tried it on Power BI.

ITEM 8 · The dimension that serves both

about 15 minutes

What to do

- 28. Look at your drawing. Find the dimension or dimensions that connect to BOTH fact tables.
- 29. Name them.
- 30. Now imagine you had built two separate copies — one for Sales, one for Inventory.

ANALYSIS

The name of the shared dimension, and the two fact tables it serves.

Shared dimensions	Connects to	Join key	Used for
tblCalendar	tblSales, tblInventory	Date	Time and date
tblBranches	tblSales, tblInventory	BranchID	Branch comparison
tblProducts	tblSales, tblInventory	ProductID	Product analysis

REFLECTION

Describe one specific report that would go wrong if Sales and Inventory each had their own private copy of that dimension. Be concrete: name the report, and say what number would be wrong.

One report that could go wrong is a branch sales and inventory report. If Sales and Inventory had separate copies of the Branch dimension, the branch names and totals could be grouped differently, which could give the wrong comparison between sales revenue and inventory movements.

B3 — FORMAL SUBMISSION BLOCK

Copy your final answers here. This is the page marked against the syllabus milestone. Everything in it should already exist somewhere in Items 1 to 8 — if it does not, go back.

1 • Grain statement, one per fact table

Written as a full sentence. "One row of ... is ..."

One row of tblSales is a single item within a sale, recorded with its SaleID, Date, Branch, Product, etc.

Total rows: 10,589

Distinct values: 10,566

The two counts are different. This means some SaleID values appear more than once. After checking actual rows, the rows show information about a specific product or item sold, such as the product, quantity, price, and amount.

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2 • Fact tables, with their measures

Column	Bucket	Why
SaleID*	DESCRIPTIVE ATTRIBUTE	Identifies the sale but is not something to add or average.
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Amount	MEASURE	Shows the sale amount and can be added to get total revenue.
PaymentMethod	DESCRIPTIVE ATTRIBUTE	Used to group or filter sales by payment method.

3 · Dimension tables, with the key each one joins on

Table	Fact / Dimension / Neither	Grain (facts only)	Join key (dimensions only)
tblSales	FACT	One row is one product or item in a sale.	
tblInventory	FACT	One row is one stock movement for a product at a branch on a date.	
tblCustomers	DIMENSION		CustomerID
tblBranches	DIMENSION		BranchID
tblCalendar	DIMENSION		Date
tblProducts	DIMENSION		ProductID
tblEmployees	NEITHER		

4 · Relationships — one line each, in the form DIMENSION 1 — * FACT (key)

tblCalendar 1 — * tblSales (Date)

tblCalendar 1 — * tblInventory (Date)

tblBranches 1 — * tblSales (BranchID)

tblBranches 1 — * tblInventory (BranchID)

tblProducts 1 — * tblSales (ProductID)

tblProducts 1 — * tblInventory (ProductID)

tblCustomers 1 — * tblSales (CustomerID)

5 · One design decision you are not certain about, and what you would need to resolve it

One design decision I am not certain about is whether Sales and Inventory should use the same Product and Date dimensions or have separate versions.

Schema drawing attached / photographed: YES Name: Sanchez, Eliah Therrien Estrella

How this is marked

Criterion	Weight
Grain statements are correct, complete and written as sentences	25
Facts, dimensions and keys classified correctly, with the hard cases defended	25
Schema drawing — complete, keyed, sides marked	20
Reflections — honest, specific, and about consequence rather than restatement	25
Submitted complete and on time	5

An item with a correct analysis and an empty or restating reflection earns roughly half of what it could. That is deliberate. The analysis shows you can operate the tool; the reflection shows you understand what you just did.

— Sir JB